Jiawei Lian, Curriculum Vitae

lianjiawei@mail.nwpu.edu.cn +86-15978793013
Google Scholar Research Gate Github

-**Birth Date**: 01/1997
-**Status**: Second-year Ph.D. Student, Northwestern Polytechnical University
-**Research Interest**: Trustworthy machine learning, Remote sensing, Computer vision

-**Postal Address**: No. 1 Dongxiang Road, Chang'an District, Xi'an City, China (710129)

Education

04/2022 – Present	Ph.D., Information and Communication Engineering, Advisor: Shaohui Mei Northwestern Polytechnical University
09/2019 – 03/2022	M.E., Control Engineering, Advisor: Junhong He Northwestern Polytechnical University Thesis: *Research on Rapid Detection of Surface Defects Based on Deep Learning*
09/2015 – 07/2019	B.E., Automation, Advisor: Junlin Zhu JiangXi University of Science and Technology Thesis: *Design of 4R Joint Robot Control System based on PROFIBUS and T-CPU*

Research Publications

Journal Articles

1. S. Mei, **J. Lian**, X. Wang, Y. Su, M. Ma, and L.-P. Chau, "A comprehensive study on the robustness of image classification and object detection in remote sensing: Surveying and benchmarking," *arXiv preprint arXiv:2306.12111*, 2023. (**Submitted to Proceedings of the IEEE, the first author is the doctoral advisor**).
2. **J. Lian**, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," ***IEEE Transactions on Geoscience and Remote Sensing***, vol. 61, pp. 1–16, 2023.
3. **J. Lian**, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," ***IEEE Transactions on Geoscience and Remote Sensing***, vol. 60, pp. 1–16, 2022.
4. **J. Lian**, J. He, Y. Niu, and T. Wang, "Fast and accurate detection of surface defect based on improved yolov4," ***Assembly Automation***, vol. 42, no. 1, pp. 134–146, 2021.
5. **J. Lian**, J. He, Y. Niu, and T. Wang, "Rapid detection of surface defects based on multi-scale compression cnn," ***Computer Integrated Manufacturing System***, vol. 28, no. 11, p. 3624, 2021.

Conference Articles

1. **J. Lian**, X. Wang, Y. Lu, M. Ma, and S. Mei, "**********for anonymous review**********," in *AAAI*, 2024. **(Submitted to AAAI 2024)**.
2. **J. Lian**, X. Wang, Y. Su, M. Ma, and S. Mei, "Contextual adversarial attack against aerial detection in the physical world," in *arXiv preprint arXiv:2302.13487*, 2023. **(International Geoscience and Remote Sensing Symposium 2023, accepted)**.

China Patents

16/05/2023 — A method and device for generating adversarial camouflage patterns against object detection. (Patent acceptance number: 202310547602.8)

16/12/2022 — A method and device for physical attack-oriented self-adaptive adversarial patch generation. (Patent acceptance number: 202211617496.8)

15/12/2022 — A method and device for generating adversarial patches for unobstructed physical adversarial attacks. (Patent acceptance number: 202211612346.8)

05/06/2021 — A fast surface defect detection method based on the lightweight convolutional neural network. (Patent acceptance number: 202110627893.2)

Miscellaneous Experience

Awards

12/2017 — Jiangxi Provincial College Students' Science and Technology Innovation and Vocational Skills Competition, Electronic Comprehensive Design Competition, First Prize of Undergraduate Group.

08/2017 — Jiangxi Provincial College Students' Science and Technology Innovation and Vocational Skills Competition, Electronic Thematic Design Competition, Second Prize of Undergraduate Group.

Certifications

04/2022 — **International English Language Testing System (IELTS): Overall band score 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0)**

07/2017 — **Fitness Coach Licentiate (FCL): Awarded by The Ministry of Human Resource and Social Security, The People's Republic of China.**

Volunteer Works

03/2022 — Volunteer in Northwestern Polytechnical University's COVID-19 epidemic prevention and control volunteer service.

07/2019 — Participate in the "Guarding the Qinling Mountains, College Students in Action" public welfare and environmental protection activity.

Involved Projects

09/2022-02/2023 — Research on the robustness of remote sensing image classification and object detection model based on deep learning.

12/2021-08/2022 — Research on cross-domain lossless evolution of adversarial perturbations for physical attack. **Relevant research achievements received an interview invitation from the British authoritative magazine New Scientist**.

Internship

07/2021-09/2021 — Internship at AVIC Xi'an Flight Automatic Control Research Institute.

07/2020-09/2020 — Internship at Xi'an Institute of Modern Control Technology.